Status	Finished
Started	Tuesday, 20 October 2026, 11:45 AM
Completed	Tuesday, 20 October 2026, 11:49 AM
Duration	3 mins 58 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Linear search ✓

c. Hash search

d. Binary search

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The search continues in the right sublist

c. The list is sorted

d. The search ends successfully ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist

b. The list must be sorted

c. Number values should only be present

Number values should only be present

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Quick search
- b. Bubble search
- c. Merge search
- d. Linear search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the right sublist ✓
- b. In the middle of the list
- c. At the beginning of the list
- d. In the left sublist

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 99 ✓
- b. 89 and 99
- c. 90 and 94
- d. 89 and 94

Question **7**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 10 ✓
- b. 1000
- c. 500
- d. 100

Question **8**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Binary search ✓
- b. Hash search
- c. Sequential search
- d. Linear search

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It does not work with negative numbers
- b. It is slower than linear search for small lists
- c. It can only be applied to large lists
- d. It requires the list to be sorted ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Dynamic search
- b. Linear search ✓
- c. Binary search
- d. Hashing

Question **11**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(n log n)
- b. O(1)
- c. O(log n) ✓
- d. O👎

Question **12**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and sorted
b. When the list is large and unsorted
c. When the list is small and unsorted ✓
d. When the list is large and sorted

Question **13**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. log n
c. n-1
d. n ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓
b. The first element is returned
c. The last element is returned
d. An error is raised

Question **15**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)
b. Hash
c. Linear
d. Binary ✓